

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 11A.4

RATE EQUATIONS AND MECHANISMS

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 11: KINETICS 2
SUB-TOPIC:	11A.4: Rate Equations and Mechanisms
TOTAL QUESTIONS:	4 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	4 Marks
EXAMINER FAQs:	3 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 11A.4

**Q1. Q1. The rate equation for a reaction is $\text{rate} = k[\text{A}][\text{B}]^2$.
Suggest the species involved in the rate-determining step.**

[1 Marks]**WCH14/01/Jun22/Q5(a)**

Q2. Q2. Define the term "rate-determining step".

[1 Marks]**WCH14/01/Jun22/Q5(b)**

**Q3. Q3. The reaction between propanone and iodine in acidic conditions is zero order with respect to iodine.
Explain what this tells you about the mechanism.**

[1 Marks]**WCH14/01/Jan21/Q2(a)**

**Q4. Q4. SN1 and SN2 mechanisms have different kinetics.
Which mechanism is associated with a first-order overall rate equation?**

[1 Marks]**WCH14/01/Jan21/Q2(b)**

SECTION B: WORKED MARK SCHEME — 11A.4

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: Q1. The rate equation for a reaction is $\text{rate} = k[\text{A}][\text{B}]^2$. Suggest the species involved in the rate-determining step. • One molecule of A and two molecules of B (1)	[1]
Q2	Q2: Q2. Define the term "rate-determining step". • The slowest step in a multi-step reaction mechanism (1)	[1]
Q3	Q3: Q3. The reaction between propanone and iodine in acidic conditions is zero order with respect to iodine. Explain what this tells you about the mechanism. • Iodine is not involved in the rate-determining step / is involved after the rate-determining step (1)	[1]
Q4	Q4: Q4. SN1 and SN2 mechanisms have different kinetics. Which mechanism is associated with a first-order overall rate equation? • SN1 mechanism (1)	[1]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 11A.4**FAQ #1: Can a reaction have a termolecular rate-determining step?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Can a reaction have a termolecular rate-determining step?

MODEL ANSWER / EXAMINER EXPECTATION:

It is very rare as the probability of three species colliding simultaneously with the correct orientation and sufficient energy is extremely low.

FAQ #2: Does the rate-determining step always occur first?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Does the rate-determining step always occur first?

MODEL ANSWER / EXAMINER EXPECTATION:

No, it can occur at any point in the mechanism. Intermediates formed before the RDS will appear in the rate equation.

FAQ #3: How do SN1 and SN2 differ in their RDS?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

How do SN1 and SN2 differ in their RDS?

MODEL ANSWER / EXAMINER EXPECTATION:

SN1 RDS involves only the halogenoalkane (unimolecular), while SN2 RDS involves both the halogenoalkane and the nucleophile (bimolecular).